



Strathmore
UNIVERSITY

STRATHMORE INSTITUTE OF MATHEMATICAL SCIENCES
MASTER OF SCIENCE IN DATA SCIENCE & ANALYTICS

CAT 1- Open Book
DSA 8505: Bayesian Statistics

DATE: 14th Feb 2025

Instruction

(a) Answer All Question

(b) Scan and submit your answer sheet through the Google Classroom by 23h59, 14th Feb 2025.

- ✓ 1. Consider a Bayesian estimation problem where the posterior distribution of a parameter θ follows a normal distribution:

$$\theta|x \sim N(\mu, \sigma^2).$$

Construct a 90% credible interval for θ using the equal-tailed credible interval method. chapter 2

- * Ques. check ✓ 2. A biased die has an unknown probability θ of landing on a six. A Bayesian statistician assumes a Beta(3,3) prior for θ . The die is rolled 15 times, and a six appears 6 times. Compute the posterior distribution of θ (Start by deriving posterior distribution). # Pm = per

- * loss function ✓ 3. The posterior distribution of a parameter θ is given by:

since dens is normal
⇒ mode = median

$$\pi(\theta|x) \sim N(10, 16).$$

$$L(\theta) = (\theta - a)^2$$

(a) Determine the Bayes estimator under the squared error loss function. chapter 3 $\hat{\theta}_{Bayes} = E(\theta|x)$

(b) Determine the Bayes estimator under the absolute error loss function. mode = 10

- ✓ 4. Suppose the posterior distribution of θ has two peaks at $\theta = 2$ and $\theta = 6$, with the highest posterior density occurring at $\theta = 6$.

(a) Explain why the MAP estimate under the 0-1 loss function differs from the Bayes estimator under squared error loss.

(b) If the posterior mean is 4.5, which estimate would you choose under squared error loss, and why?